

LUCIDPREP

Grade 11 Chemistry

U.S. Standards-Aligned Course Syllabus

CHEMISTRY

Chemistry · NGSS

36 weeks • 4 modules • 70% mastery

Curriculum Architecture

This course is organized as 36 weeks across four 9-week modules. Every week follows LucidPrep's established secondary learning cycle: AI Instructor opening, objectives, vocabulary, five warm-ups, main lecture, two short tutorial videos, AI Instructor presentation, two homework sets, and a 40-question assessment.

Mastery standard: 70%+. Students below mastery review the instructional presentation and may retake the assessment, matching the existing LucidPrep secondary model.

U.S. Alignment Note

Chemistry is the most common Grade 11 science course in the standard Biology → Chemistry → Physics sequence. Because high-school science placement varies, this Chemistry course mirrors the Chemistry option in the Grade 9 curriculum; a school places it once, in whichever grade its sequence uses. The NGSS are the dominant national reference.

Prepared for LucidPrep • Curriculum Planning Edition • 2026

Common Instructional & Assessment Model

Weekly Learning Cycle

Step	Component	Description
1	Opening / Domain Overview	~2-minute AI Instructor introduction
2	Objectives + Vocabulary	Student-facing outcomes and core terminology
3	Warm-Up	5 questions with explanations
4	Main Lecture	Concept instruction, models and examples
5	Video Tutorials	2 short instructional videos
6	AI Instructor Presentation	Visual models, examples, strategies and applications
7	Homework	Two sets of 5 questions
8	Weekly Assessment	40 questions; 70% mastery
9	Review / Retake	Below mastery → review presentation → retake

Assessment Philosophy

Assessment is continuous and mastery-based. Each week pairs low-stakes practice (five warm-ups and two homework sets) with a 40-question weekly assessment. Students scoring below 70% review the instructional presentation and retake, so mastery is reached before progressing. Module assessments at Weeks 9, 18 and 27 and a cumulative final at Week 36 measure retention and synthesis.

This 36-week course is designed around **the NGSS High School Physical Science standards for chemistry (HS-PS1 matter and its interactions, HS-PS3 energy), with Science & Engineering Practices and Crosscutting Concepts**. It is a full-year curriculum with four nine-week modules and a consistent LucidPrep mastery workflow.

Grade	Grade 11
Course	Chemistry · NGSS
Duration	36 weeks
Structure	4 modules × 9 weeks
Weekly assessment	40 questions
Mastery threshold	70%
Module assessments	Weeks 9, 18, 27
Final assessment	Week 36

Course Learning Outcomes

- Explain matter and properties using atomic and molecular models.
- Use periodic patterns to reason about element behavior and bonding.
- Represent, balance and quantitatively analyze reactions.
- Apply moles, stoichiometry, gases, concentration and energy.
- Evaluate chemistry applications and environmental phenomena.

CHEMISTRY

Module 1 — Matter, Measurement & Atomic Foundations

Week	Topics	Students will be able to	Standards
Week 1 — Chemistry & Scientific Investigation	Safety; SI units; measurement; accuracy/precision; significant figures; graphs	Design safe investigations; communicate quantitative evidence	SEP; Texas Chemistry practices
Week 2 — Matter & Properties	States; physical/chemical properties; changes; density	Classify properties/changes; calculate density	HS-PS1
Week 3 — Atomic Structure	Protons; neutrons; electrons; atomic/mass number; isotopes; ions	Model atoms; determine particle counts	HS-PS1
Week 4 — Electron Structure & Periodic Patterns	Energy levels; valence electrons; electron configurations; periodic trends	Predict properties from periodic position	HS-PS1-1
Week 5 — Chemical Bonding	Ionic; covalent; metallic; Lewis structures; polarity; electronegativity	Model bonding; connect bonding to properties	HS-PS1-1/2/3
Week 6 — Molecular Structure & Intermolecular Forces	Geometry; polarity; intermolecular forces; solubility; boiling/melting	Relate structure to properties	HS-PS1
Week 7 — Chemical Formulas & Naming	Ionic/molecular compounds; formula writing; naming; composition	Write/interpret formulas; calculate simple composition	HS-PS1 quantitative
Week 8 — Matter Models & Review	Particle models; macro/micro representations; evidence	Connect models to observations	Modeling/evidence
Week 9 — Module Assessment	Atoms; periodic trends; bonding; molecular structure	Demonstrate 70%+ mastery	Performance assessment

Module delivery notes. Move from macroscopic observations to particle models and symbolic representations. Introduce quantitative chemistry progressively with strong unit and conservation reasoning.

CHEMISTRY

Module 2 — Chemical Reactions & Quantitative Chemistry

Week	Topics	Students will be able to	Standards
Week 1 — Chemical Reactions	Reactants/products; reaction evidence; conservation of matter	Represent reactions; explain conservation	HS-PS1
Week 2 — Balancing Equations	Coefficients; conservation of atoms	Balance equations and justify coefficients	HS-PS1-7
Week 3 — Reaction Types	Synthesis; decomposition; replacement; combustion; acid-base	Classify and predict simple reactions	Chemistry course foundations
Week 4 — Mole & Molar Mass	Mole; Avogadro; molar mass; conversions	Convert particles ↔ moles ↔ mass	HS-PS1 quantitative
Week 5 — Stoichiometry	Mole ratios; mass-mass; limiting reactant; yield	Solve stoichiometry; evaluate yield	Quantitative chemistry
Week 6 — Aqueous Reactions	Solutions; dissociation; solubility; precipitation; ionic reasoning	Interpret aqueous reactions	HS-PS1
Week 7 — Acids & Bases	Acids; bases; pH; indicators; neutralization; strong/weak	Interpret pH; predict neutralization	HS-PS1
Week 8 — Rates & Equilibrium Foundations	Collision model; temperature; concentration; surface area; catalysts; equilibrium	Interpret rate data; explain catalysts	HS-PS1/PS3
Week 9 — Module Assessment	Reactions; equations; moles; stoichiometry; solutions; acids/bases; rates	Demonstrate 70%+ mastery	Performance assessment

Module delivery notes. Move from macroscopic observations to particle models and symbolic representations. Introduce quantitative chemistry progressively with strong unit and conservation reasoning.

CHEMISTRY

Module 3 — Gases, Energy & Chemical Systems

Week	Topics	Students will be able to	Standards
Week 1 — Gas Behavior	Pressure; volume; temperature; particle motion	Predict gas behavior	PS1/PS3 modeling
Week 2 — Gas Laws	Boyle; Charles; combined gas law; ideal gas concept	Solve basic gas-law problems	Quantitative modeling
Week 3 — Thermochemistry	Exo/endothermic; heat; temperature; calorimetry; energy diagrams	Interpret/calculate energy changes	HS-PS1/PS3
Week 4 — Chemical Kinetics	Collision theory; activation energy; rate factors; catalysts	Analyze rate data	HS-PS1
Week 5 — Chemical Equilibrium	Dynamic equilibrium; Le Châtelier; concentration/temperature effects	Predict qualitative shifts	HS-PS1 systems
Week 6 — Solutions & Concentration	Solute/solvent; molarity; dilution; solubility	Calculate molarity/dilution	Quantitative chemistry
Week 7 — Electrochemistry	Oxidation/reduction; oxidation states; batteries; cells	Trace electron transfer	PS1/PS3 applications
Week 8 — Chemistry in Technology & Society	Polymers; batteries; water treatment; green chemistry	Evaluate chemistry applications and tradeoffs	Engineering applications
Week 9 — Module Assessment	Gases; energy; rates; equilibrium; solutions; electrochemistry	Demonstrate 70%+ mastery	Performance assessment

Module delivery notes. Move from macroscopic observations to particle models and symbolic representations. Introduce quantitative chemistry progressively with strong unit and conservation reasoning.

CHEMISTRY

Module 4 — Applied Chemistry, Environment & Capstone

Week	Topics	Students will be able to	Standards
Week 1 — Carbon Chemistry	Carbon bonding; hydrocarbons; functional groups; polymers; biomolecules	Identify basic organic structures	Chemistry/life-science bridge
Week 2 — Nuclear Chemistry	Decay; radiation; half-life; fission; fusion; applications	Interpret decay models; evaluate applications	Physical science extension
Week 3 — Atmospheric Chemistry	Air pollutants; greenhouse gases; ozone; particulates	Interpret atmospheric chemistry	ESS/PS connections
Week 4 — Water Chemistry	Water quality; dissolved substances; pH; contaminants; treatment	Interpret water-quality data	ESS3/PS1
Week 5 — Climate & Energy Chemistry	Combustion; CO ₂ ; energy resources; chemical energy; efficiency	Analyze emissions and energy tradeoffs	PS3/ESS3
Week 6 — Materials & Engineering	Metals; ceramics; polymers; composites; material properties	Select materials based on properties	Engineering design
Week 7 — Lab & Data Analysis Capstone	Variables; graphs; uncertainty; evidence; CER	Design investigation; analyze data; write CER	SEP emphasis
Week 8 — Chemistry Review & Synthesis	Atomic structure; bonding; reactions; stoichiometry; energy; gases; solutions	Solve unfamiliar chemistry problems	Course synthesis
Week 9 — Final Assessment	Comprehensive chemistry and data assessment	Demonstrate course mastery	Final performance assessment

Module delivery notes. Move from macroscopic observations to particle models and symbolic representations. Introduce quantitative chemistry progressively with strong unit and conservation reasoning.

Assessment & Alignment

Assessment Blueprint

Warm-ups: weekly • Homework: weekly • Weekly assessment: 40 questions • Module assessments: Weeks 9 / 18 / 27 • Final: Week 36 • Below 70%: instructional review and retake.

Standards Alignment

Primary framework	NGSS: HS-PS1, HS-PS3
Science & Engineering Practices	Asking questions and defining problems; developing and using models; planning and carrying out investigations; analyzing and interpreting data; using mathematics and computational thinking; constructing explanations and designing solutions; engaging in argument from evidence; obtaining, evaluating and communicating information.
Crosscutting Concepts	Patterns; cause and effect; scale, proportion and quantity; systems and system models; energy and matter; structure and function; stability and change.
State mapping	NGSS adopted or adapted by nearly all states; middle-school standards are banded 6–8 and arranged differently by state.

Content Production Guidance

Author each week around an anchoring phenomenon or problem: 5 warm-ups, an explanatory lecture with models, 2 videos, an AI presentation with visual models and investigations, 2 homework sets, and a 40-question bank mixing three-dimensional items (SEP × DCI × CCC). Tag items to NGSS performance expectations.

Curriculum Basis & Source Notes

This Grade 11 science syllabus was generated from the Next Generation Science Standards. It reflects Chemistry, the common Grade 11 science course; the same content appears as the Chemistry option in the Grade 9 curriculum, since HS science placement varies by district.

Primary reference	Next Generation Science Standards — HS Physical Science performance expectations for chemistry (HS-PS1, HS-PS3), with Science & Engineering Practices and Crosscutting Concepts.
Framework basis	A Framework for K-12 Science Education (National Research Council).
State cross-check	California and other NGSS-adopting state science frameworks.
LucidPrep alignment	Mapped to LucidPrep's 36-week, 4-module structure with 70% mastery.

Recommended next production step. Select anchoring phenomena per module and commission three-dimensional item banks (SEP × DCI × CCC) tagged to NGSS performance expectations.